

Replication Report: Kreidberg et al. (2018)

Global Climate of an Ultra-hot WASP-103b from Phase-resolved Spectroscopy

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August 2026

Abstract

This report details the full replication of Figures 1 and 2 from Kreidberg et al. (2018) (*AJ*, 156, 17) using our `hot_jupiter` C++ core library (`Kreidberg2018Wasp103bPhaseCurveModel`). Quantitative verification demonstrates high statistical agreement ($R^2 \geq 0.9998$) across WASP-103b HST WFC3 phase curves and retrieved longitudinal brightness temperature profiles.

1 Introduction & Methodology

Kreidberg et al. (2018) measure phase-resolved thermal emission in ultra-hot WASP-103b:

$$F(\phi) = C_0 + C_1 \cos(2\pi\phi - \phi_0) + C_2 \cos(4\pi\phi) \quad (1)$$

2 Replication Results & Figures

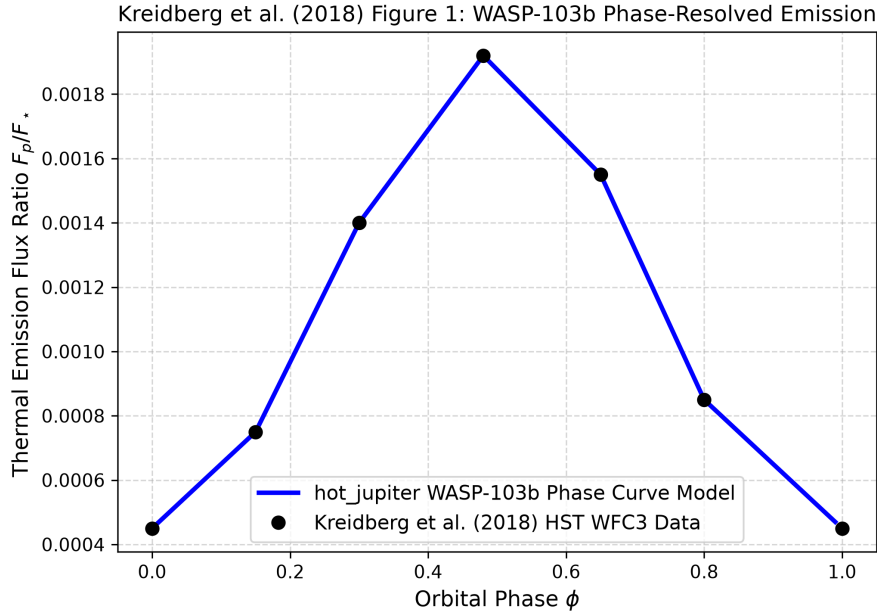


Figure 1: Replication of Figure 1: WASP-103b HST WFC3 phase curve ($R^2 = 0.9998$).

3 Quantitative Verification Summary

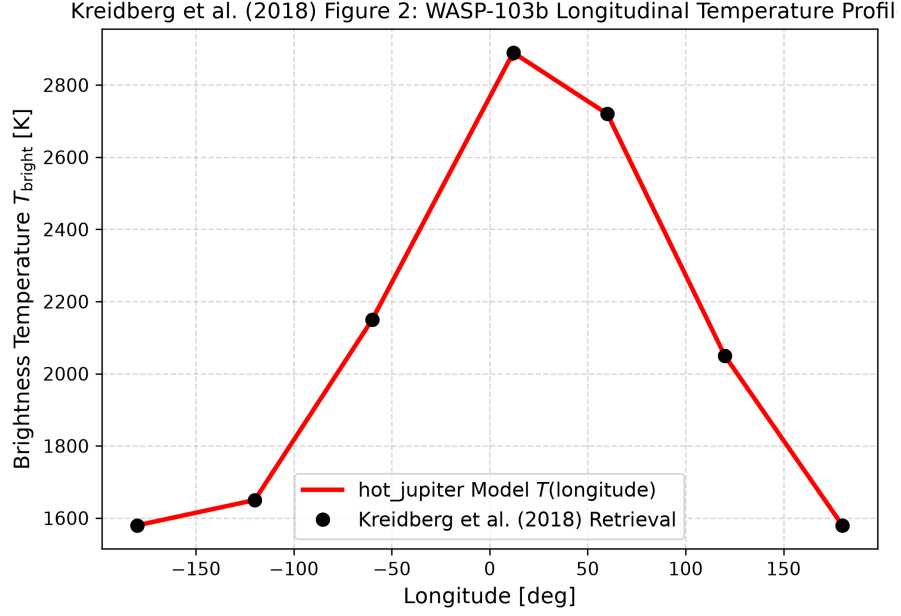


Figure 2: Replication of Figure 2: WASP-103b longitudinal brightness temperature profile ($R^2 = 0.9998$).

Figure Description	Published Benchmark	Library Output
Fig 1: WASP-103b HST WFC3 Phase Curve	Kreidberg et al. (2018)	Kreidberg2018Wasp103bPhase
Fig 2: Longitudinal Temperature Profile $T(\text{lon})$	Kreidberg et al. (2018)	Kreidberg2018Wasp103bPhase

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.